Variable importance of the closing-interval models

Bullets of Section 6.5 (parked, note E19)

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- **Models:** the closing-interval models of record on the full predictor set ($p = 628$), trained on the last 30 minutes of each trading day (Section G.5), not on the intraday scheme.
  - Lasso and ridge with the penalty strength λ selected on a pre-sample: $\lambda = 10^{-2}$ and 10^{3} (scikit-learn parametrization). Penalty weight ω_j, the factor on predictor j's penalty term: $\underline{\omega} = 0.01$ on the HAR benchmark block, 1 on the exogenous predictors. Re-estimated every trading day; the scheduled re-selection of λ never replaced the pre-sample choice.
  - Not the earlier study's single-penalty specification (one λ for every predictor, re-selected every 250 trading days): the paper does not use it.
  - Tree ensembles at fixed hyperparameters, the best entries of the development-sample search (LightGBM and XGBoost: the top Optuna trial; random forest: trial 255); constant and duplicate predictors dropped in each estimation window.
- **Notation.**
  - Re-estimation $k = 1, \ldots, K$, $K = 147$, one every 10 trading days. Re-estimation k uses the window $\mathcal{W}_{t_k}$, the 2,000 trading days before its first forecast day t_k, and forecasts its out-of-sample block $\mathcal{O}_k$, the trading days from t_k to the next re-estimation (≤ 10). The blocks make up this section's forecast days $\mathcal{O} = \bigcup_k \mathcal{O}_k$: $|\mathcal{O}| = 1{,}469$ trading days, 25 June 2018 to 30 April 2024.
  - Predictors $\boldsymbol{x}_t \in \mathbb{R}^p$; forecast $\hat{y}_t = \hat{f}_k(\boldsymbol{x}_t)$, $t \in \mathcal{O}_k$, on the scale of the target y_t, the diurnally adjusted realized volatility of the closing half hour.
  - Predictor group $A \subseteq \{1, \ldots, p\}$: all predictors built from one series (its averages at every horizon, its missing-data and activity indicators). 49 groups: 47 exogenous series in 8 categories (Section 4), the HAR components, the calendar dummies.
  - Linear models: $\hat{\boldsymbol{\beta}}_k \in \mathbb{R}^p$, the coefficient estimates behind the forecast for t_k; $\hat{\boldsymbol{\beta}}_{k,A}$, its subvector on group A.
  - Trees: SHAP values $\phi_{t,j}$ (Lundberg and Lee, 2017) and SHAP interaction values $\phi_{t,jl}$ (Lundberg et al., 2020); they add up to the forecast,

$$\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}, \qquad \phi_{t,j} = \sum_{l=1}^{p} \phi_{t,jl}, \qquad \phi_{t,A} = \sum_{j \in A} \phi_{t,j}.$$

    For $j \neq l$, $\phi_{t,jl} = \frac{1}{2}(\phi_{t,l} \mid j \text{ present} - \phi_{t,l} \mid j \text{ absent})$, and $\phi_{t,jj} = \phi_{t,j} - \sum_{l \neq j} \phi_{t,jl}$.
  - Path-dependent TreeSHAP (Lundberg et al., 2020): the training observations per node (node covers) stand in for the background sample.
  - Number triples below: LightGBM, XGBoost, random forest.

Table 1: Variable importance by model

Panel A: Linear models

	Lasso		Ridge		Ridge, standardized	
Rank	Group	K_A	Group	S_A (%)	Group	S_A^{sd} (%)
1	Calendar dummies	147	Calendar dummies	31.0	HAR components	30.6
2	HAR components	147	HAR components	22.8	Calendar dummies	9.3
3	`sumret`	147	`sumret`	3.6	`sumret`	4.5
4	`sumabsret_vwstock`	147	`sumabsret_vwstock`	2.2	`sumabsret_vwstock`	3.7
5	`sumautocov`	147	`sumautocov`	2.1	`effspread_vwstock`	2.5
Kendall's W		0.74		0.85		–
Series above probe		23		38		–

Panel B: Tree ensembles

	LightGBM		XGBoost		Random forest	
Rank	Group	$\bar{r}_A$	Group	$\bar{r}_A$	Group	$\bar{r}_A$
1	HAR components	1.0	HAR components	1.0	HAR components	1.0
2	`sumret`	2.9	`sumret`	3.0	`sumabsret`	2.6
3	`sumabsret`	4.8	`sumabsret`	3.9	`sumret`	4.0
4	`sumvolume`	4.9	`sumvolume`	4.6	`sumvolume`	5.1
5	`sumabsret_ewstock`	6.7	`sumabsret_ewstock`	6.7	`sumabsret_ewstock`	5.7
Kendall's W		0.57		0.60		0.62
Series above probe		27		35		29

Notes: Closing-interval models on the full predictor set ($p = 628$ predictors in 49 predictor groups), $K = 147$ re-estimations, one every 10 trading days, over the out-of-sample period 25 June 2018 to 30 April 2024 ($|\mathcal{O}| = 1{,}469$ trading days). Each panel lists the five leading groups by the measure in the column header (Section 6.5): the lasso's selection count K_A (re-estimations with a nonzero coefficient on group A, of K; ties broken by the lasso's coefficient share), the ridge's share S_A of the absolute coefficients $\sum_k \sum_j |\hat{\beta}_{k,j}|$ and its share S_A^{sd} of the absolute standardized coefficients $\sum_k \sum_j |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}$, and the tree ensembles' mean rank $\bar{r}_A$ by mean absolute SHAP value (1 = most important). Kendall's W is the concordance of the K rankings, one per re-estimation ($W = 1$: the same ranking at every re-estimation). Series above probe: exogenous series with a predictor that beats the continuous noise probe ε^{U}, added in 49 probe fits (lasso: selected in a larger share of re-estimations than the probe in its fits; ridge and trees: a larger $|\hat{\beta}_{k,j}|$ or mean $|\phi_{t,j}|$ than the probe's in at least half of the probe fits). Group symbols as in Table 5.

- **Summary table:** Table 1 ranks the predictor groups model by model, as in the variable-importance rankings of Gu et al. (2020), each model by its own aggregate measure below: the lasso's selection count K_A, the ridge's coefficient shares S_A and S_A^{sd}, and the trees' mean SHAP rank $\bar{r}_A$. Under each column: Kendall's W of the rankings across re-estimations, and the number of exogenous series with a predictor that beats the random probe.
- **Lasso, aggregate:** the selection count of a group and of a predictor, the number of the K re-estimations with a nonzero coefficient,

$$K_A = \sum_{k=1}^{K} \mathbf{1}\{\exists j \in A : \ \hat{\beta}_{k,j} \neq 0\}, \qquad K_j = \sum_{k=1}^{K} \mathbf{1}\{\hat{\beta}_{k,j} \neq 0\};$$

  K_A/K is the selection frequency, the share of estimation windows with a nonzero coefficient.
  - Per re-estimation: 30 nonzero coefficients (median; range 27–35), of which 14 on exogenous predictors.

  - Predictors: 60 of 628 selected at least once, 29 in at least half of the re-estimations.
  - Exogenous series: 27 of 47 selected at least once; 11 in at least half of the re-estimations; 3 in every re-estimation (`sumret`, `sumabsret_vwstock`, `sumautocov`).
  - HAR benchmark block: selected in every re-estimation ($\omega_j = 0.01$); 88 % of $\sum_j |\hat{\beta}_{k,j}|$ (median re-estimation).
  - Random-probe benchmark: the continuous probe, an i.i.d. uniform noise predictor (defined under Known issues), is selected in 2.0 % of its fits; 54 predictors, in 23 exogenous series, are selected more often.
- **Lasso, time-varying:** the magnitude and the selection indicator of each group at each re-estimation,

$$\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 = \sum_{j \in A} |\hat{\beta}_{k,j}|, \qquad \mathbf{1}\{\|\hat{\boldsymbol{\beta}}_{k,A}\|_1 > 0\}$$

  Rank stability across re-estimations: Kendall's coefficient of concordance W of the rankings by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ at each re-estimation ($W = 1$: the same ranking at every re-estimation; formula below), $W = 0.74$.
- **Ridge, aggregate:** the group's share of the absolute coefficient mass,

$$S_A = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}|}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}|}.$$

  - Scale check: only the averages of series values are robust-scaled.
    * Averages of series values (282 predictors): standard deviation over the estimation window 0.84 (median; 5th–95th percentiles: 0.03–1.76), interquartile range 0.94.
    * Averaged missing-data and activity indicators (324): standard deviation 0.022; calendar dummies 0.39; HAR components 1.07.
    * So the raw $|\hat{\beta}_{k,j}|$ are in different units. Check with the group's share of the absolute standardized coefficients,

$$S_A^{\mathrm{sd}} = \frac{\sum_k \sum_{j \in A} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}}{\sum_k \sum_{j=1}^{p} |\hat{\beta}_{k,j}| \, \mathrm{sd}_{k,j}},$$

      with $\mathrm{sd}_{k,j}$ the standard deviation of predictor j over $\mathcal{W}_{t_k}$; Spearman rank correlation of S_A^{sd} with S_A across groups 0.85.
    * Leading groups on that scale: HAR components 30.6 %, calendar dummies 9.3 %, `sumret` 4.5 %.
  - Leading groups: calendar dummies 31.0 %, HAR components 22.8 %, `sumret` 3.6 %; HAR benchmark block 54 % (median re-estimation).
  - Ridge sets no coefficient to zero: all 49 groups have $K_A = K$, so the selection frequency does not discriminate.
  - Random-probe benchmark: 107 predictors, in 38 exogenous series, have $|\hat{\beta}_{k,j}|$ above the continuous probe's (robust-scaled) in most probe fits.
- **Ridge, time-varying:** the share at re-estimation k, $S_{k,A} = \|\hat{\boldsymbol{\beta}}_{k,A}\|_1 / \sum_{A'} \|\hat{\boldsymbol{\beta}}_{k,A'}\|_1$, A' over all groups (mean rank by $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$); Kendall's $W = 0.85$.
- **Trees, ranking at each re-estimation:** the group's mean absolute SHAP value on the out-of-sample block of re-estimation k, then its rank,

$$I_{k,A} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,A}|, \qquad r_{k,A} = 1 + \#\{A' : I_{k,A'} > I_{k,A}\} \quad \text{(ties averaged)}.$$

- **Trees, global ranking:** aggregate the K rankings over time,

$$\bar{r}_A = \frac{1}{K}\sum_{k=1}^{K} r_{k,A}, \qquad \mathrm{BC}_A = \sum_{k=1}^{K} \max(0,\ 11 - r_{k,A}).$$

  - BC_A: truncated Borda count (a full Borda count is affine in $\bar{r}_A$).
  - Kendall's coefficient of concordance (Kendall and Babington Smith, 1939) over the K rankings of the n groups, with rank sums $R_A = \sum_k r_{k,A}$ and their mean $\bar{R}$:

$$W = \frac{12\sum_A (R_A - \bar{R})^2}{K^2(n^3 - n) - K\sum_k \sum_{g\in\mathcal{E}_k} (|g|^3 - |g|)},$$

    $\mathcal{E}_k$ the sets of tied (equally ranked) groups at re-estimation k; $W = 1$: the same ranking at every re-estimation.
  - LightGBM: HAR components ($\bar{r}_A = 1.0$), `sumret` (2.9), `sumabsret` (4.8); Kendall's $W = 0.57$.
  - XGBoost: HAR components (1.0), `sumret` (3.0), `sumabsret` (3.9); Kendall's $W = 0.60$.
  - Random forest: HAR components (1.0), `sumabsret` (2.6), `sumret` (4.0); Kendall's $W = 0.62$.
  - Mean rank vs truncated Borda count, Spearman rank correlation over groups: 0.90, 0.87, 0.92.
  - Aggregate of the ranks vs rank of the aggregate, the mean absolute SHAP value over $\mathcal{O}$, $\bar{I}_A = |\mathcal{O}|^{-1}\sum_{t\in\mathcal{O}} |\phi_{t,A}|$: 0.96, 0.97, 0.98.
  - Predictor level ($n = 628$): Kendall's $W = 0.91$, 0.90, 0.93.
  - Agreement of $\bar{r}_A$ across models (Spearman rank correlation): trees 0.83–0.96; tree vs linear 0.17–0.55; lasso vs ridge 0.47.
- **Interaction effects, global:** the mean absolute SHAP interaction value of two groups over $\mathcal{O}$,

$$\phi_{t,AB} = \sum_{j\in A}\sum_{l\in B} (\phi_{t,jl} + \phi_{t,lj}), \qquad \bar{I}_{AB} = \frac{1}{|\mathcal{O}|}\sum_{t\in\mathcal{O}} |\phi_{t,AB}|, \quad A \neq B;$$

  for two predictors, $|\phi_{t,jl} + \phi_{t,lj}|$, $j \neq l$.
  - Top pair: LightGBM HAR components × `sumabsret` (12.6); XGBoost HAR components × `sumabsret` (15.2); random forest HAR components × `sumabsret` (23.8); $\times 10^{-3}$, in units of y_t.
  - Scale: largest $\bar{I}_A$ 195, 192, 224 ($\times 10^{-3}$); all series-pair $\bar{I}_{AB}$ summed, as % of $\sum_A \bar{I}_A$: 90, 74, 72.
  - Within a group ($\sum_{j\neq l\in A} \phi_{t,jl}$): largest HAR components (18.7), HAR components (19.1), HAR components (34.3); $\times 10^{-3}$.
  - Noise floor: the continuous probe's largest $\bar{I}$ with any group, median over probe fits: 0.49, 0.41, 0.50 ($\times 10^{-3}$); series pairs above it: 116, 115, 52 of 1,176.
- **Interaction effects, cross-check:** the H-statistic of Friedman and Popescu (2008) on each model's top 10 series pairs, from the partial dependence functions PD_S, $S \in \{A, B, AB\}$, $AB \equiv A \cup B$:

$$H^2_{AB} = \frac{\sum_t [\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)]^2}{\sum_t \mathrm{PD}_{AB}(\boldsymbol{x}_t)^2}, \qquad \mathrm{PD}_S(\boldsymbol{x}) = \frac{1}{|\mathcal{H}_k|}\sum_{t'\in\mathcal{H}_k} \hat{f}_k(\boldsymbol{x}_S, \boldsymbol{x}_{t',-S}),$$

  $t \in \mathcal{H}_k$; each PD_S centered over $\mathcal{H}_k$.

  - Points: $\mathcal{H}_k$, the 100 trading days from t_k on, both evaluation points and background sample, at 15 re-estimations (every tenth); sums pooled over re-estimations.
  - The denominator contains the HAR components' own partial dependence whenever they are in the pair. A second normalization, by the smaller single-group partial dependence:

$$H^2_{\min,AB} = \frac{\sum_t \left[\mathrm{PD}_{AB}(\boldsymbol{x}_t) - \mathrm{PD}_A(\boldsymbol{x}_t) - \mathrm{PD}_B(\boldsymbol{x}_t)\right]^2}{\min\{\sum_t \mathrm{PD}_A(\boldsymbol{x}_t)^2,\ \sum_t \mathrm{PD}_B(\boldsymbol{x}_t)^2\}}.$$

  - Top pair, H^2 (%): 0.33, 0.37, 0.85; over the top 10: 0.02–0.33, 0.02–0.68, 0.02–0.85.
  - Top pair, $H^2_{\min}$: 0.19, 0.39, 1.50; over the top 10: 0.05–0.54, 0.08–0.80, 0.17–5.46.
  - Top predictor pair, H^2 (%) and $H^2_{\min}$: LightGBM 0.36, 0.02; XGBoost 0.39, 0.02; random forest 1.16, 0.06.
- **Interaction effects, time-varying:** $\phi_{t,AB}$ on the out-of-sample block of each re-estimation, ranked at each re-estimation and aggregated as the SHAP values are,

$$I_{k,AB} = \frac{1}{|\mathcal{O}_k|} \sum_{t \in \mathcal{O}_k} |\phi_{t,AB}| \ \rightarrow\ r_{k,AB} \ \rightarrow\ \bar{r}_{AB},\ W.$$

  - Kendall's W over each model's top 50 pairs: 0.45, 0.47, 0.57.
  - Mean rank vs $\bar{I}_{AB}$ (Spearman rank correlation, all pairs): 0.99, 0.99, 1.00.
- **Known issues, reconciled:**
  - Mean decrease in impurity (MDI) favors predictors with many distinct values (Strobl et al., 2007): MDI is not used; SHAP values and coefficients only.
  - Random probes, at every third re-estimation (49 fits): three i.i.d. noise predictors, continuous (U(0, 1)), discrete uniform on 20 levels, and binary (Bernoulli, probability one half).
    - Lasso: the probes are selected in 1, 0, 0 of 49 fits (robust-scaled, $\omega_j = 1$).
    - Ridge: 107 predictors have $|\hat{\beta}_{k,j}|$ above the continuous probe's in most probe fits.
    - Trees: the continuous probe's median rank among 631 predictors by mean $|\phi_{t,j}|$: 83, 110, 97; exogenous series with a predictor above it in most probe fits: 27, 35, 29.
  - Dilution across correlated predictors: $\phi_{t,A}$ and $\|\hat{\boldsymbol{\beta}}_{k,A}\|_1$ sum over the group's predictors, so credit split among a series' averages at different horizons stays inside the group.
  - The earlier study's single-penalty lasso (λ re-selected): median 23 nonzero coefficients per re-estimation; 18 exogenous series selected in at least half of the re-estimations; Spearman rank correlation of its K_A with that of the lasso of record 0.70.
  - Reproduction: re-running the estimation reproduces the archived LightGBM models exactly; XGBoost and the random forest differ across library versions (maximum absolute forecast difference 0.17 and 0.11, in units of y_t). SHAP values: from the archived models; interaction values: from the reproduced ones (Spearman rank correlation of $\bar{r}_A$, reproduced vs archived: 1.000, 0.993, 0.996).
- **Compute** (one laptop, 11 processes):
  - Reproducing the tree models and computing the SHAP interaction values: 133 minutes. Per re-estimation: random forest estimation 212 s, its SHAP and interaction values 182 s; LightGBM 0.9 s, XGBoost 0.4 s.
  - Linear coefficient paths and probes: 7 minutes; H^2: 7 minutes.

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